

Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-Informed Neural-Network Optimization Framework — Final Presentation

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PHYS 400 — Final Presentation





Outline

1

Motivation

2

Method — five-stage pipeline

3

Results

4

Discussion & summary



Every load-bearing part is sized by two numbers:

- $E = \sigma/\varepsilon$ — Young's modulus: stiffness (GPa).
- $\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$ — Poisson's ratio: transverse contraction.

For cubic crystals both follow from the two stiffness constants C_{11} and C_{12} — **the quantities this work targets from first principles.**

They set: beam deflection $\delta \propto 1/E$, buckling load $\propto E$, natural frequency $\propto \sqrt{E/\rho}$; bulk modulus $K = E/[3(1 - 2\nu)]$, stress concentration, toughness.

Stability (cubic, Born–Cauchy): $C_{11} > C_{12}$.

Representative structural applications

Application	Material	E (GPa)	ν
Aerospace fuselage	Al Alloys	60–80	0.3–0.38
Turbine disk	Inconel 718	200	0.29
Orthopaedic im-plant	Ti 6Al–4V	110	0.34
Automotive structure	Mg AZ91	45	0.35
Structural steel	Mild steel	200	0.30
MEMS resonator	Si (single xtal)	130	0.28

Same two constants span polymers ($\nu \rightarrow 0.4$) to steel — and fix whether a design survives load.



Interatomic potential. Closed-form approximation of the many-body potential energy $U(\{r_i\})$; forces obtained as $F_i = -\nabla_i U$.

Historical development:

- Pairwise (Lennard–Jones, Morse) — too crude for metals.
- Embedded Atom Method (1984) — good for FCC metals.
- **Modified EAM (MEAM)** [1]: adds angular terms — covers BCC, HCP and mixed bonding.

Density-functional theory (DFT).

First-principles electronic-structure method (Kohn–Sham equations [2]); here a plane-wave + pseudopotential basis in Quantum Espresso [3].

Quantities computed:

- Cohesive energy E_{coh}
- Equilibrium lattice a_0 , bulk modulus B
- Binary formation energies $E_{\text{form}}(i, j)$
- Single-crystal elastic constants C_{11} , C_{12}

DFT: high accuracy, limited scale ($\leq 10^3$ atoms).

MEAM/MD: empirical, large scale ($\geq 10^6$ atoms).

DFT provides the reference dataset against which the MEAM potential is calibrated.

Why No Existing Potential Suffices: The MEAM Coverage Gap



Published MEAM libraries cover disjoint subsets of elements; no single file spans the nine elements specified by Al7075.

Library	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	
Merged (this work)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Limitation of direct merging: cross-interaction parameters between elements drawn from distinct libraries are not defined; manual refitting does not scale to $\binom{12}{2} = 66$ binary pairs.

Pure machine-learning potentials [4–6]: require $10^4 - 10^5$ DFT configurations and forfeit the physical interpretability of MEAM.

02

Method — five-stage pipeline



1

Motivation

2

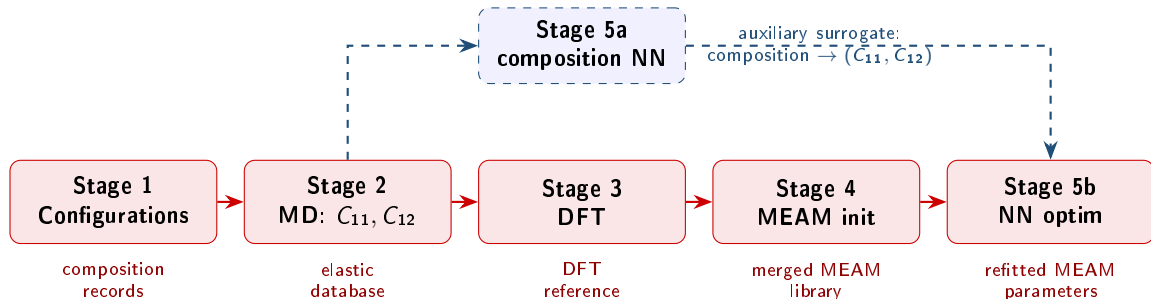
Method — five-stage pipeline

3

Results

4

Discussion & summary



The four published libraries are merged into a single 12-element file; cross-terms are refitted by inverting an NN surrogate of the MEAM $\rightarrow (C_{11}, C_{12})$ map by gradient descent, with single-element parameters anchored to DFT references.



Sampling.

Uniform draws on the 12-element composition simplex

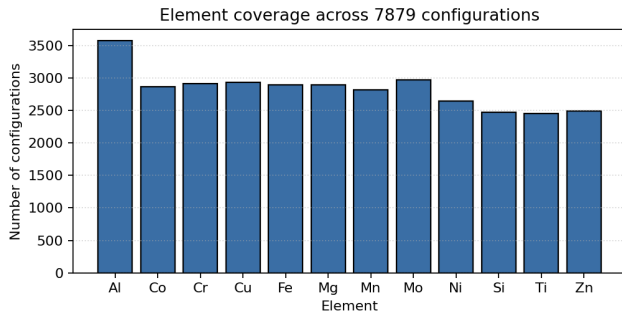
(Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn);

Al-weighted to oversample the aluminium-alloy region.

Corpus statistics.

~7900 configurations generated;

7815 retained after the physicality filter
($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).



Element-occurrence distribution over the retained corpus.



Implementation. [7]

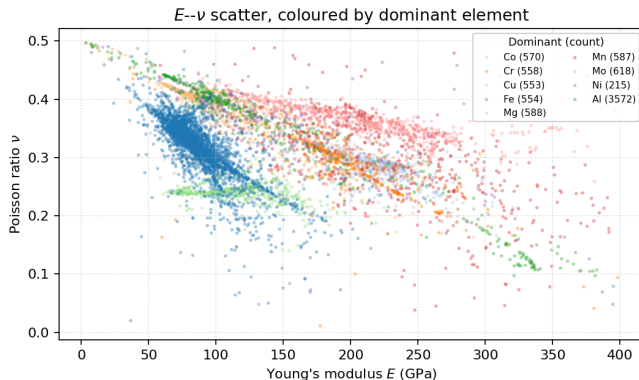
LAMMPS molecular dynamics with the merged 12-element MEAM potential.

Per-configuration protocol:

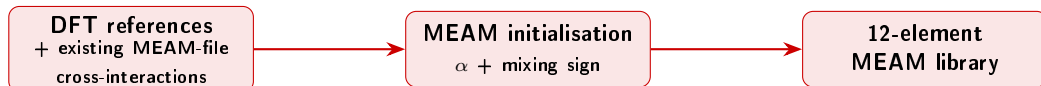
- ~ 4 nm cubic supercell
- Anisotropic energy minimisation
- Axial strain $\delta = 10^{-3}$ applied along $\pm x, \pm y, \pm z$
- Central-difference $\rightarrow C_{11}, C_{12}$

	Median	Std.	Range
E (GPa)	103.6	69.1	3 – 398
ν	0.326	0.063	0.01 – 0.50

$N = 7815$ retained configurations.



E - ν distribution, coloured by dominant element.



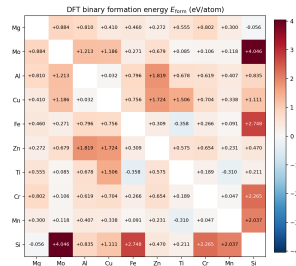
Single-element terms

Each element's (a_0, E_{coh}, B) pins the universal binding curve — spacing, well depth, curvature — fixing the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{\text{coh}}} \quad (\Omega : \text{atomic volume}).$$

Scope. 12 elements (FCC/BCC/HCP/diamond); $\alpha \approx 4.5$ (Al, Si) to 10.9 (Zn). PBE-GGA plane waves (Quantum Espresso); a_0, B from an equation-of-state fit.

Cross-terms



Binary formation energies $E_{\text{form}}(i, j)$ (eV/atom): negative = exothermic mixing, positive = limited solubility — the sign seeds each cross-term.

03

Results



1

Motivation

2

Method — five-stage pipeline

3

Results

4

Discussion & summary

Stage 5a: Composition $\rightarrow (E, \nu)$ Neural Surrogate

TensorFlow / Keras

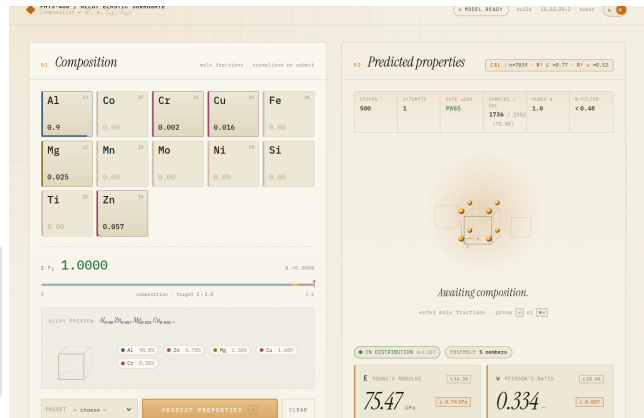


Architecture. $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$
fully-connected MLP
with ReLU activation; Huber loss on $\hat{C}_{11}, \hat{C}_{12}$;
deep ensemble of five seeds [8] for
epistemic uncertainty.

Result

Validation Set (30 % of the dataset):
 $R_E^2 = 0.77$, $R_\nu^2 = 0.52$.

Al7075 Example: $\hat{E} = 75.5$ GPa, $\hat{\nu} = 0.334$
(published: 71–72 GPa, $\nu \approx 0.33$ — within $\sim 5\%$).



Interactive composition predictor.



Closed-loop training process

- 1 **Surrogate.** A 4-layer MLP ($\rightarrow 20 \rightarrow 20 \rightarrow 10 \rightarrow$, ReLU) learns (MEAM params) $\rightarrow (C_{11}, C_{12})$.
- 2 **Fit.** Trained with *Adam* [9] on a *Huber* loss ($\delta = 1$), up to 10^4 epochs, early-stopped once the loss falls below 5×10^{-3} .
- 3 **Inverse step.** Adam (lr 0.01, 3000 steps) descends through the *frozen* surrogate to the MEAM parameters that hit the target (C_{11}, C_{12}).
- 4 **Loop.** Optimised parameters are re-evaluated in LAMMPS; new samples are appended and the surrogate retrained (active learning).



Surrogate Huber-loss decays $1.0 \rightarrow 0.16$ over training.

Why these choices

Adam — adaptive-moment gradient optimiser: per-parameter step sizes from running gradient moments, a robust default for NN training.

Huber loss — squared-error for small residuals, absolute-error for large ones; outlier configurations (Mg/Zn) cannot dominate the fit.

Result — a working proof of concept

Tested on 30 unseen alloys:

- 11 gave a physically stable potential;
- 4 matched the MD reference within $\sim 20\%$.

The pipeline runs end-to-end; accuracy grows with more data and longer training (next slide).



By material family ($N = 43$)

Family	N	E MAPE (%)		ν MAPE (%)	
		ML	MEAM	ML	MEAM
Aluminium	33	2.3	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Ni-Fe (A-286)	1	107.0	—	—	—
Overall	43	10.8	42.2	7.2	7.1

Aluminium by AA series ($N = 33$)

Al series	N	$\langle E_{lit} \rangle$ (GPa)	E MAPE (%)	
			ML	MEAM
2xxx (Al-Cu)	11	72.7	1.3	8.0
5xxx (Al-Mg)	9	70.5	2.1	7.9
6xxx (Al-Mg-Si)	7	68.9	3.8	4.2
7xxx (Al-Zn-Mg)	6	71.8	2.8	9.3
All aluminium	33	71.1	2.3	7.4

Both surrogates reproduce aluminium Young's modulus to within a few percent (ML 2.3 %, MEAM 7.4 %) across all four alloy series; error grows only *off* the Al-rich design domain — steeply for MEAM, gently for the composition NN.

43 basis-covered alloys; MAPE vs experimental data from ASM/MatWeb [10]. Unphysical MEAM runs excluded.

04

Discussion & summary



1

Motivation

2

Method — five-stage pipeline

3

Results

4

Discussion & summary



Achieved

- ✓ **Automated 12-element MEAM pipeline** — an end-to-end workflow that assembles a multi-element potential.
- ✓ **Aluminium-alloy elastic prediction** — E , ν and C_{11} , C_{12} obtained from composition.
- ✓ **DFT-seeded, NN-trained MEAM** — first-principles references refit through a neural surrogate.

The pipeline already runs end to end; what is left is more data, longer training, and denser DFT.

What comes next

- 1 **More data coverage** — wider sampling across the composition simplex and crystal structures.
- 2 **More NN training epochs** — deeper surrogate convergence before each refit.
- 3 **Denser atomic-cell DFT** — finer k -point and supercell sampling for sharper references.



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